Jev-PHM: Non-Generative Fault Diagnosis Across Datasets with Heterogeneous Annotations

Abstract

Cross-dataset fault diagnosis requires transferring signal information without silently changing what the diagnostic labels mean. Generating a class name is not essential to this task, but removing response decoding does not resolve either measurement shift or incompatible supervision. We formulate Jev-PHM as a backbone-independent method for non-generative prediction over semantic fault candidates. The proposed method uses a bounded signal representation, shared candidate scoring, and supervision that distinguishes exact labels from verified coarsenings. Two questions guide the study: when can compatible supervision preserve distinctions on an excluded dataset, and how does direct probability readout change the diagnostic error–latency trade-off relative to label generation? An explicit annotation-map condition and a native mixed-damage counterexample establish when the proposed event likelihood is applicable; coarse labels alone still do not identify every fine-state probability. A label-token scoring control also limits the computational claim: a one-token diagnostic answer need not require a sequential response-decoding loop. The formulation is accompanied by a completed, narrower pilot on 2,559 recordings from 32 Paderborn bearings. With six temporal statistics and a frozen Qwen3-0.6B encoder, question-factorized minus direct-prediction negative log-likelihood is −0.00110, with conditional 95% interval [−0.00858, 0.00516]. Six refitted rotations change direction, and every primary calibrated Qwen method defers all observations at the stated cost. These results bound the earlier question-decomposition approach; they do not validate the new semantic scorer, cross-dataset transfer, or reduced latency. A separate physical-simulation study tests soft conditional preservation and a hard-freezing control on a fixed 46-feature representation. Source-only selection chooses a zero soft penalty and no additional update in all three cohorts; identical selected predictions do not establish algorithmic equivalence. In the implemented waveform study, source-side fitting stopped when the best semantic configuration reached its update cap without qualification, leaving external evaluation sealed.

1. Introduction

1.1 Diagnosis across measurements and label definitions

Within prognostics and health management (PHM), a fault classifier deployed on another machine or dataset encounters more than a new distribution of vibration amplitudes. Sampling, sensor configuration, operating conditions, and the definition of the target fault can all change. A label denoting predominantly inner-ring damage need not mean an isolated inner-ring defect, and a coarse bearing-fault label does not identify a particular damaged component. Physical-bearing separation is also necessary when recordings from one bearing recur across operating conditions (Lessmeier et al. 2016; Vieira et al. 2026). A shared model must preserve these distinctions rather than treat dataset-specific class numbers as interchangeable targets.

Existing fault-diagnosis models address important parts of this problem. Heterogeneous Signal Embedding (HSE) addresses heterogeneous signal inputs (Q. Li et al. 2025). UniFault learns representations from multiple bearing datasets and evaluates downstream adaptation, whereas YOTOnet evaluates transfer to excluded datasets under a unified fault vocabulary (Eldele et al. 2025; Wang et al. 2026). These settings distinguish representation reuse, target-data adaptation, and target-free transfer. None is synonymous with applying the same software separately to several datasets.

Language-based diagnosis introduces a second design choice. FaultGPT produces diagnostic answers from time-frequency representations, while HCAA uses a language-model arbitrator after a preliminary probabilistic diagnosis (Chen et al. 2025; Wu 2025). A fixed-category diagnosis can instead return class probabilities directly. This avoids response-token decoding and answer parsing by construction, but does not imply lower end-to-end latency than every numerical model, higher diagnostic accuracy, or calibrated probabilities.

1.2 Scientific questions and motivation

The first scientific question concerns supervision rather than framework construction: **under measurement shift and source labels with different but verifiably compatible granularity, when does learning their annotated events reduce fault-classification error or negative log-likelihood on an entirely excluded dataset, compared with learning a common coarse vocabulary or using only exact source labels?** The conditions are consequential. If the sources never distinguish two fault states, a normalized distribution does not identify their individual probabilities. If their annotations describe different physical events, no relabeling rule can make them equivalent without additional assumptions.

The second question concerns a computational–statistical relation: **at fixed signal information, requested fault granularity, and source-only model selection, how does semantic-candidate readout change the error–latency relation relative to exact label-token scoring and minimal label generation as token budget and candidate count vary?** The comparison holds the accepted diagnostic coverage fixed or reports the coverage–error trade-off alongside latency. A single-token language-model decision can already avoid sequential response decoding; the baseline cannot be limited to lengthy explanations. A fast predictor that rejects all observations does not establish useful diagnostic throughput. The two questions are independent: annotation compatibility can affect transfer even without language generation, while changing the readout can affect computation even when every label is exact.

Jev’s typed-decision interface motivates the separation of diagnostic probability estimation from response generation (TypeSafe AI 2026b, 2026a). GLiClass already studies generalist direct classification, and NanoJev already uses small pretrained backbones with question- and candidate-conditioned decision heads (Stepanov et al. 2025; TianyuCodings 2026). Jev-PHM therefore does not claim to invent direct classification, candidate scoring, or a small-model interface. The candidate increment is the effect of preserving native fault-label distinctions in shared semantic learning, not another implementation of candidate scoring. The readout comparison separately measures computation at the same diagnostic task and accepted coverage. The method is not defined by one backbone family. An individual model uses one trained checkpoint across datasets; a different backbone requires a separate instantiation and training configuration.

1.3 Contributions and current support

We formulate semantic fault-candidate prediction with exact and coarsened supervision, and state the condition under which the source annotation is determined by the chosen diagnostic state. Applying this condition to the documented Paderborn mixed-damage bearings rules out two tempting but invalid label mappings. This source-specific applicability result constrains the proposed likelihood; it is not a new general learning theorem. The candidate learning rule retains source distinctions rather than fabricating fine labels. Whether it reduces excluded-dataset risk beyond conventional semantic learning remains untested.

The completed empirical contribution is a controlled pilot separating diagnostic question conditioning from ordinary hierarchy and an additional direct prediction. Six comparison arms receive the same six temporal statistics. The primary paired interval includes zero, a hierarchy on the direct language representation has a

lower point loss, and all three Qwen contrasts reverse direction across six rotations. Lower expected calibration error (ECE) also coexists with a higher area under the risk–coverage curve (AURC), and every calibrated primary Qwen arm rejects every test window. These observations support a bounded negative finding for the pilot, not the effectiveness of the proposed semantic method.

A separate simulation compares ordinary event likelihood, the released weighted Libra loss, a source-selected soft conditional penalty and hard conditional freezing on the same measured features and initialization. The selected soft workflow performs no additional update, so a regularization effect along an unselected trial cannot explain its final diagnostic score. This adds a controlled qualification of treatment activation, not evidence of a superior preservation algorithm. The manuscript separates the general formulation, completed real-data pilot and simulation study. Real cross-dataset transfer, semantic-backbone portability, and the error–latency relation still require the comparisons stated in Section 7. The first scope is single-label fault diagnosis over a declared, complete candidate set, not remaining useful life prediction, multi-label fault localization, unseen-fault detection, or validated maintenance execution.

2. Related Work

2.1 Transferable fault representations and diagnostic interfaces

HSE addresses heterogeneous signal inputs (Q. Li et al. 2025). UniFault v2 combines multi-dataset pretraining with target-task adaptation (Eldele et al. 2025). Its experiment passages differ in wording: one describes exclusion of fine-tuning samples, whereas another states exclusion of all samples from the three adapted datasets. We do not resolve that provenance ambiguity by assuming either contamination or verified whole-dataset exclusion. Target-label adaptation is explicit and remains different from target-free deployment. YOTOnet instead holds out datasets, but studies an inner-ring/outer-ring binary fault task and reports optional target-labeled adaptation separately (Wang et al. 2026). Discriminating two known fault types does not evaluate healthy-versus-faulty detection or mixed-damage recognition. These are distinct estimands, not interchangeable levels of a single benchmark. Our pending comparison must therefore match both target-data access and the physical meaning of the requested classes.

FaultGPT uses vision-language instruction supervision, and HCAA supplies a preliminary diagnosis, features, and diagnostic images to an arbitrator (Chen et al. 2025; Wu 2025). Their outputs and supervision differ from shared non-generative candidate prediction; their published scores are not same-protocol baselines. CLSGen jointly trains classification and explanation-generation heads, whereas the completed pilot uses a frozen encoder (Yoon et al. 2026). A discriminative language-model head is established prior art.

2.2 Non-generative semantic prediction

CLIP supplies the observation–text compatibility principle (Radford et al. 2021). GLiClass uses joint text/label encoding as its main design and separately specifies a bi-encoder with shared scoring (Stepanov et al. 2025). Candidate precomputation follows the latter design; it cannot inherit the label interactions or measured results of the joint encoder automatically. Our dot-product readout belongs to this established family. Using the same architecture, supervision and optimization would reproduce an ordinary semantic baseline rather than introduce a different algorithm. The proposed comparison changes which verified annotation information is retained, while keeping the readout fixed.

NanoJev supplies direct probability heads (TianyuCodings 2026). Jung studies Jev on heating, ventilation, and air-conditioning (HVAC) faults, comparing independent decisions with language models instructed to output one word; label probabilities are recovered from returned next-token alternatives (Jung 2026). Thus

short-answer controls already exist in Jev diagnosis. Our evaluation requires exact scores for every candidate token rather than a truncated top-token list, and separates source-only calibration from target calibration. The HVAC results do not supply vibration-transfer or matched-latency evidence. These precedents rule out a broad first-Jev-diagnosis claim.

2.3 Partial labels, coarsened labels, and incompatible faults

Partial-label learning supplies a set containing one correct class and studies when that ambiguity can be resolved (Cour, Sapp, and Taskar 2011). Coarse-label learning instead formalizes observations induced by partitions of a finer label space and requires sufficient information preservation for recovery (Fotakis et al. 2021). We use the latter interpretation only for a verified deterministic annotation partition. We do not reproduce Cour et al.'s convex surrogate or inherit Fotakis et al.'s statistical-query guarantee for a neural empirical optimizer under domain shift.

A mixed-damage observation is not uncertainty about which single pure defect occurred. Likewise, a dominant-damage annotation need not be a coarsening of pure single-fault labels. These distinctions decide whether source targets can share a probability space at all. The proposed contribution, if validated, would be the effect of preserving genuine PHM annotation semantics during transfer, not the invention of a set-likelihood expression. Fault-presence and fault-identity targets derived from a single diagnosis are also not independently measured physical concepts of the kind supervised in concept bottleneck models (Koh et al. 2020).

Predictive-retention mechanisms are also established. Libra loss studies allowed-label probability ratios under explicit architectural assumptions (Zombori et al. 2024). Learning without Forgetting (LwF) distils an earlier predictor on new inputs, while Orthogonal Gradient Descent (OGD) projects updates using stored output gradients (Z. Li and Hoiem 2018; Farajtabar et al. 2020). Their soft or first-order preservation mechanisms cannot be equated with exact global conditional freezing, but they rule out claiming that distillation or projection itself is new. Our simulation tests whether such control survives fair source-side selection; it does not transfer their guarantees to a new semantic backbone.

2.4 Calibration, selective prediction, and independent evaluation

Temperature scaling, conditional classifiers, and rejection are established methods (Guo et al. 2017; Valmadre 2022; Chow 1970; Geifman and El-Yaniv 2017). Hierarchical selective classification can vary prediction specificity, but all comparisons here must use the same requested label granularity (Goren, Galil, and El-Yaniv 2024). A coarse answer cannot be credited as a correct fine diagnosis only for the proposed method. CAT combines calibration with teacher–student domain adaptation in fault diagnosis, rather than the pilot's fixed-classifier post-hoc calibration (Forest and Fink 2024).

Probability losses, binned calibration error, and confidence ordering evaluate different properties (Gneiting and Raftery 2007; Cattelan and Silva 2024). Their joint reporting is especially important when low scenario cost can be obtained by always deferring. Physical-unit grouping follows the Paderborn dataset's bearing identities and the concerns established in bearing-level evaluation research (Lessmeier et al. 2016; Vieira et al. 2026). More overlapping windows do not create more independent equipment, and target adaptation must not be mislabeled as generalization.

2.5 Comparison of supervision, retention, and validation scope

Table 1 compares eight concrete methods at their stated level of analysis. Direct-classification and diagnostic-transfer methods establish the input/output and evaluation precedents; loss and continual-learning methods establish the relevant retention mechanisms. The table is not a performance ranking across incompatible tasks. Source-specific definitions and original-method locations are provided in the comparison notes.

Table 1. Supervision, prediction mechanisms, and validation scope of related methods. Symbols indicate whether a dimension is addressed, restricted or not established; they do not indicate that a method is superior.

Work / version	Supervision	Semantic candidates	Retention mechanism	Shift evaluation	Probability assessment	Inference measurement
GLiClass (2025, v1) (Stepanov et al. 2025)	Exact single-/multi-label	✓ Label descriptions	—	○ Text-task transfer	—	✓ GPU throughput
UniFault (2025, v2) (Eldele et al. 2025)	Self-supervised; target fine-tuning	✗ Indexed target head	—	○ Target-label adaptation	—	—
YOTOnet (2026, v1) (Wang et al. 2026)	Common exact fault labels	✗ Indexed fault head	—	✓ Excluded datasets; two faults	—	—
Jev–HVAC (2026, v1) (Jung 2026)	Inference study; pretraining undisclosed	✓ Textual alternatives	—	✓ Seasonal/system/building	✓ Brier, ECE	✓ Hosted/local timing
CAT (2024) (Forest and Fink 2024)	Labelled source; unlabelled target	✗ Fixed class head	—	○ Target-data adaptation	✓ Calibration/ECE	—
Libra loss (2024) (Zombori et al. 2024)	Sets containing one true label	✗ Indexed softmax	○ Allowed-label ratio property	—	○ Probability dynamics	—
LwF (2018) (Z. Li and Hoiem 2018)	Exact new-task labels; old predictions	✗ Task-specific heads	✓ Soft output distillation	○ Sequential image tasks	—	—
OGD (2020) (Farajtabar et al. 2020)	Exact sequential-task labels	✗ Task-specific outputs	✓ First-order output projection	○ Sequential image tasks	—	—
Ours (evaluated scope)	**Exact/coarsened simulated labels**	**○ Design only**	**○ Soft/hard tested; selected soft off**	**○ Simulated target exclusion**	**✓ NLL, Brier**	**—**

✓ denotes addressed coverage in the stated scope; ○ denotes restricted, adjacent or design-only coverage; ✗ denotes absence from the specified evaluated architecture; — denotes not established in the inspected source. Task qualifiers are essential: sequential learning and target adaptation are not target-free diagnosis, and synthetic coarsening is not industrial annotation provenance. CAT's EMA teacher is not a demonstrated fixed fine-conditional retention control. The LaTeX source separates method families and the final row with horizontal rules.

Jev–HVAC reports inference measurements, including hosted-request latency and local-model timing (Jung 2026, sec. 4.7). They must not be marked unreported, although the different execution conditions do not isolate a matched-hardware readout effect. The Ours row concerns the simulation of Section 7.4; the earlier PU study evaluates a different fixed-statistic predictor. Neither experiment implements the full semantic learner of Section 4.

Specific gap. Direct prediction from semantic candidates, diagnostic transfer, probability calibration and predictive retention each have established methods. GLiClass and Jev–HVAC provide direct-decision and timing precedents; UniFault, YOTOnet and CAT study distinct target-access regimes. Libra controls allowed-label probability dynamics, while LwF and OGD limit changes in previous predictions through distillation or projection. These results do not directly establish whether preserving a within-fault conditional distribution improves excluded-domain diagnostic risk when a new source supplies only a verified coarse event. That effect must be separated from added source information, ordinary event likelihood, model-selection behavior and a simpler conditional-freezing head. Our current contribution is a scoped qualification of this question: an explicit annotation condition and controlled tests that distinguish a mathematically defined or trial-active mechanism from one actually selected for prediction. The present simulation selects no nonzero soft-preservation treatment, so it cannot support a claim of semantic-transfer or runtime superiority.

3. Problem Formulation and Notation

3.1 Observations, source annotations, and target exclusion

An observation is a vibration window $x \in \mathbb{R}^{n \times C_x}$ and its available measurement metadata m. The positive integers n and C_x denote time samples and measured channels. Dataset d supplies an annotation y_d whose meaning is established from its documentation, not from its numerical code. A canonical finite set $\mathcal{Y} = \{1, ..., K\}$ contains mutually exclusive diagnostic states, and $\mathcal{C} \subseteq \mathcal{Y}$ is the declared candidate set for a particular classification task. Candidate descriptions are t_k, with $k \in \mathcal{C}$. Aliases for one event are not additional classes.

The initial closed-set assumption is that the true state belongs to $\mathcal{C}$. Removing a state from the denominator is justified only by the declared task population, not by absence of its annotation in one dataset. Missing labels are not known negative classes. A target dataset is excluded from task-specific training, preprocessing fits, prompt refinement from its samples, checkpoint selection, calibration, and threshold selection. Public class definitions may specify the task; target observations may not tune it. The same exclusion must be checked for domain-specific pretraining. Disjoint training and test windows, or excluding only fine-tuning samples, does not exclude the target collection. Unknown general-backbone pretraining exposure is disclosed rather than treated as verified absence.

A *backbone-independent method* has a common mathematical interface across compatible backbone families. It does not imply transferable weights between architectures. *Cross-dataset generalization* means evaluation on a dataset excluded as above. *Adaptation* means use of target data for updates or selection and is reported

separately. *Non-generative readout* means no autoregressive response-token decoding at diagnosis time; a causal-attention backbone may still be used.

Term or symbol	Definition and scope	Distinction
x, m	Measured signal and available metadata	Not a predicted equipment condition
$s(x) \in \mathbb{R}^6$	Six temporal statistics used only in the pilot	Not a sufficient signal representation
$\mathcal{C}, t_k$	Closed-set candidates and their descriptions	Not dataset-specific class IDs
$g_d, S_d(y_d)$	Deterministic annotation map and its preimage event	A coarse event is a set of exclusive diagnostic states, not a claim that only one physical lesion exists
$h, e_k \in \mathbb{R}^{d_h}$	Signal and candidate representations	d_h is embedding width, not fault probability
$p_\Theta, \tilde{p}_T$	Model probabilities before/after post-hoc temperature scaling	Maximum probability is a score, not a safety guarantee
τ, T	Training score scale and post-hoc calibration temperature	Neither is inference latency
$v_k,\ \ell_{v_k}$	Distinct single-token label code and its exact next-token logit, used only in the readout control	A code is not a fault description; top-token truncation is not exact scoring
Λ, c_{def}	Action-cost matrix and deferral cost	Separate from regularization parameter C_{reg}
Acceptance coverage	Weighted fraction of accepted diagnoses	Not a prediction-set coverage guarantee

3.2 What annotation compatibility means

For the likelihood case, dataset d has a known deterministic map $g_d: \mathcal{C} \to \mathcal{L}_d$ from diagnostic states to its annotation vocabulary. Define

$$S_d(y_d) = \{k \in \mathcal{C}: g_d(k) = y_d\}.$$

These sets form a disjoint partition of the relevant candidate population. An exact annotation is a singleton. A coarse annotation denotes the union of its compatible states. The source map is used to interpret supervision, not as a dataset-ID input to the predictor. If the annotator chooses among overlapping descriptions using an unmodeled selection mechanism, or a set only weakly contains the true label, this deterministic likelihood interpretation does not hold. Such data require a stated alternative weak-supervision model and are excluded from the initial likelihood experiment rather than silently repaired.

A diagnostic state may include co-occurring lesions while remaining exclusive of other *complete states*. The partition assumption concerns those states, not mutual exclusivity of individual damaged components. To check whether the state definition is sufficient, let Ω be admissible physical states, $\pi: \Omega \to \mathcal{C}$ their canonicalization, and $a_d: \Omega \to \mathcal{L}_d$ the deterministic source annotation rule. Then

$$\exists g_d: \pi(\Omega) \to \mathcal{L}_d, \quad a_d = g_d \circ \pi \quad \Leftrightarrow \quad \pi(\omega_1) = \pi(\omega_2) \Rightarrow a_d(\omega_1) = a_d(\omega_2).$$

Proof. Necessity follows by applying g_d to equal canonical states. For sufficiency, define $g_d(k) = a_d(\omega)$ for any preimage with $\pi(\omega) = k$. Constancy on each preimage makes this definition independent of the choice, and hence $a_d = g_d \circ \pi$. States outside $\pi(\Omega)$ are not identified by the argument. This elementary factorization condition establishes applicability, not a statistical generalization bound.

The distinction is concrete in Paderborn: KB23 and KB24 have both-ring damage with inner-ring dominance, whereas KB27 has both-ring damage with outer-ring dominance (Lessmeier et al. 2016). The retained cohort contains 240 recordings from these three bearings. A vocabulary limited to health and pure single-ring damage does not cover them. Adding one undifferentiated both-ring state still merges different annotations, violating the displayed condition. Putting that state into both the inner-dominant and outer-dominant compatibility sets gives annotation probabilities summing to $1 + p_{\text{both}}$, where p_{both} is its predicted probability in the four-state vocabulary. Those overlapping events are not the assumed partition. A dominance-refined state space or an explicitly dominant-location task requires separate semantic justification. No pilot record is removed or relabeled. The source-specific verification is retained in the annotation analysis.

Canonicalization does not assert invariant conditional signal–fault relationships. Informative annotations and transferable measurement relationships remain separate requirements; valid source maps alone do not identify target posteriors.

3.3 Risk, acceptance, and evaluation units

Negative log-likelihood (NLL), measured with natural logarithms in nats, is the main probability endpoint. The completed compact-state pilot uses B test bearings with n_b exported windows each:

$$R_{\text{unit}}(p) = \frac{1}{B}\sum_{b=1}^{B}\frac{1}{n_b}\sum_{i=1}^{n_b} -\log p\left(y_{bi} \mid x_{bi}\right).$$

Here i flattens recordings and windows within a bearing. This is an equal-bearing mean of window losses, not a claim that windows are independent. The implemented waveform-transfer protocol in Section 4.5 instead gives the R_b original recordings within each declared unit equal weight. With n_{br} windows in recording r, its primary estimand is

$$R_{\text{record/unit}}(p) = \frac{1}{B}\sum_{b=1}^{B}\frac{1}{R_b}\sum_{r=1}^{R_b}\frac{1}{n_{br}}\sum_{i=1}^{n_{br}} -\log p\left(y_{bri} \mid x_{bri}\right).$$

These estimands differ when recording lengths vary and are not pooled. In the transfer study, PU units are documented bearing identities; MFPT units follow the explicitly assumed experiment-to-bearing convention described below. A fine-label test requires independently available fine ground truth; evaluation against a coarse label can assess only its corresponding event probability. Risk–coverage curves, binned ECE, and realized action cost are separate endpoints, not synonyms for general reliability.

For a fixed action-cost matrix Λ, the action is

$$a^*(x) = \arg\min_{a \in \mathcal{C} \cup \{\perp\}} \sum_{k \in \mathcal{C}} \Lambda_{ak}\,\tilde{p}_T(k \mid x).$$

The pilot uses correct-diagnosis cost zero, error cost one, and $c_{\text{def}} = 0.25$ for deferral. It accepts only when $1 - \max_k \tilde{p}_T(k \mid x) < c_{\text{def}}$, and defers on a tie. These are normalized experimental costs, not maintenance economics. Deferral is an action, not an unknown-fault class. Accepted-sample error is undefined when acceptance coverage is zero; rejection does not erase the underlying classifier's ordinary prediction error.

4. Proposed Semantic Decision Method

4.1 Bounded signal encoding and shared candidate scoring

This section specifies the new method; the completed experiments in Sections 5–6 evaluate the earlier compact-state pilot, not this implementation. Let E_ϕ map (x, m) to $L \leq M$ signal tokens, already projected to the selected backbone's input width. Let f_θ denote that backbone, P its stated pooling/projection operation, and G_ψ the candidate-description encoder. All trainable parameters belong to $\Theta = (\phi, \theta, \psi)$, including those of the projections. Then

$$Z = E_\phi(x, m), \qquad h = P(f_\theta(Z)) \in \mathbb{R}^{d_h}, \qquad e_k = G_\psi(t_k) \in \mathbb{R}^{d_h},$$

$$p_\Theta(k \mid x, m, \mathcal{C}) = \frac{\exp\left(h^\top e_k/\tau\right)}{\sum_{j \in \mathcal{C}} \exp\left(h^\top e_j/\tau\right)}, \qquad \tau > 0.$$

The initial readout is a dot product, not an additional mixture, retrieval system, or generated explanation. One signal encoding supports all candidates; no dataset-specific classifier rows are fitted. Candidate embeddings can be precomputed for a fixed model and candidate definitions. Reordering distinct candidates reorders the probability vector, whereas duplicating an alias would incorrectly change its normalization and is disallowed. Changing the candidate set changes the conditional classification task and need not preserve calibration.

The signal adapter must be fixed from source development data before testing. Token-budget control does not justify undocumented downsampling, aliasing, fabricated sample rates, or discarding transients without evaluation. HSE supplies relevant prior input-alignment work, not a new component claimed here (Q. Li et al. 2025). Section 4.5 specifies the implemented first instance, including patch duration, channel handling, source-only scaling and anti-alias processing. Its implementation and source-interface checks do not establish target diagnostic efficacy. The six pilot statistics remain a distinct baseline, not a substitute for signal-interface validation.

4.2 Supervision at the recorded label granularity

For a justified deterministic coarsening, the probability of observing y_d is the sum over its preimage, giving

$$\ell_d\left(\Theta; x, y_d\right) = -\log \sum_{k \in S_d\left(y_d\right)} p_\Theta(k \mid x, m, \mathcal{C}).$$

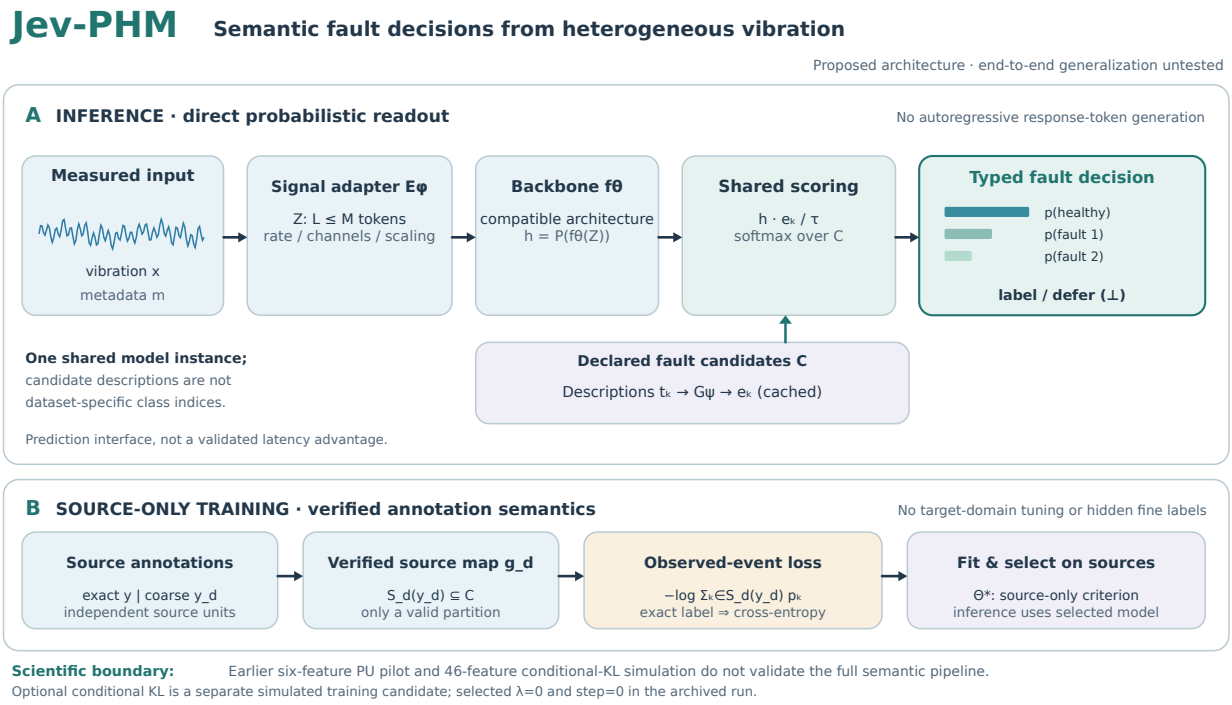
For exact labels this is ordinary cross-entropy. For sources with N_d observations and predeclared nonnegative within-source weights ω_{di} summing to one, use

$$\mathcal{L}(\Theta) = \frac{1}{D} \sum_{d=1}^{D} \sum_{i=1}^{N_d} \omega_{di}\, \ell_d\left(\Theta; x_{di}, y_{di}\right).$$

Here D is the number of training sources, not a new neural component. Weights are specified from the sampling units before fitting, not chosen from favorable outcomes; they prevent overlapping-window counts from silently setting a source's training mass. Regularization and backbone adaptation are held comparable across controls. No extra contrastive, distillation, or reinforcement-learning objective is required by the formulation.

The objective is established coarse-label likelihood, not a new generic learning guarantee (Cour, Sapp, and Taskar 2011; Fotakis et al. 2021). The informative contrast is compatible-event versus exact-label-only or common-coarse supervision using a common readout and adequate tuning. An otherwise identical semantic matcher with the same valid supervision is the same method, not an independent novelty baseline. A common-coarse predictor can be compared on aggregated event loss, but cannot be assigned fine-label NLL by inventing probabilities within its groups. If all sources have equivalent exact labels, the method reduces to conventional semantic classification and cannot support an annotation-granularity contribution. A mixed/dominant-fault label cannot be replaced by a union of pure defects unless the source semantics actually make that union correct.

The proposed Jev-PHM framework is summarized in the schematic below. Its inference pathway is separate from source-only fitting of exact or deterministically coarsened annotations. The diagram specifies a candidate method; neither the earlier six-statistic PU pilot nor the fixed-feature simulated conditional-preservation experiment is an end-to-end implementation of this semantic model.



Jev-PHM conceptual framework, not an experimental result. (A) Measured vibration and available acquisition metadata are mapped into bounded signal tokens, represented by a compatible backbone, and compared with encoded fault descriptions using shared candidate scoring. The complete categorical distribution supports a typed diagnosis or deferral without generating answer tokens. (B) Source-only training interprets observed exact or verified coarsened labels through a deterministic annotation map and fits the corresponding event likelihood, with all selection performed on source development units. The 46-feature simulation evaluated an optional conditional KL penalty but selected zero strength and no additional update; it does not validate this full framework or a latency advantage.

4.3 A concrete identifiability boundary

Suppose the only labels are healthy and faulty, while the desired fine categories are healthy, inner-ring fault, and outer-ring fault. At a fixed observation let the fault-event probability be $\rho \in (0,1)$. For every $\alpha \in [0,1]$,

$$p_\alpha = (1-\rho, \rho\alpha, \rho(1-\alpha))$$

gives exactly the same healthy/faulty probabilities. Thus coarse supervision does not identify α, even with unlimited repeated windows. More generally, when a partition matrix A maps fine probabilities to observed event probabilities, any interior p and nonzero v satisfying $Av = 0$ and $\mathbf{1}^\top v = 0$ admit sufficiently small positive and negative perturbations $p + \epsilon v$ with identical observations. This follows by substitution and is an identifiability diagnostic, not a new theorem.

Additional exact labels or informative partitions may resolve an ambiguity within a shared population, as studied in coarse-label learning (Fotakis et al. 2021). When labels and measurements both change across datasets, that transfer additionally needs shared predictive structure; concatenating source partition matrices cannot assume all sources have the same conditional posterior. Semantic names can supply an inductive bias, but do not prove the missing distinction. The proposed experiment must therefore distinguish real fine-label transfer from merely fitting the observed coarse groups.

4.4 Inference cost and decision policy

After candidate precomputation, dot-product scoring costs $O\left(|\mathcal{C}| d_h \right)$ for the declared candidates in addition to signal processing and one backbone encoding. There are zero generated response tokens. These are properties of the specified computation, not an end-to-end speedup result. Signal acquisition duration, preprocessing, warm inference, candidate compilation, and any downstream review are distinct costs. Candidate-dependent cross-encoding of the full observation would be a different algorithm with different complexity.

A necessary control is exact single-token label scoring. At the specified response boundary, let distinct tokenizer-verified tokens v_k encode the declared classes and let ℓ_{v_k} be their next-token logits after encoding the complete prompt and signal. The restricted class distribution is

$$p_{\text{token}}\left(k \mid x, m, \mathcal{C} \right) = \frac{\exp\left(\ell_{v_k} \right)}{\sum_{j \in \mathcal{C}} \exp\left(\ell_{v_j} \right)}.$$

Every candidate logit is read, including low-scoring candidates absent from a returned top-token list. This computes a decision from one prompt encoding without a sequential response-decoding loop. It conditions on the allowed token set, not on all possible generated strings, and supplies no calibration guarantee. Multi-token labels require their stated sequence-scoring rule; shared first pieces cannot be treated as distinct class probabilities. The source-chosen prompt specifies the same class meanings for both readouts, without target-driven selection of label codes.

Consequently, removing response decoding alone cannot establish a speed advantage over this control. Compare the same eligible backbone and signal information using exact token scoring, shortest valid label generation, semantic readout, and a compact non-generative signal classifier. Vary M and the candidate count $|\mathcal{C}|$ using source-declared task settings; report error or NLL jointly with measured p50/p95 latency, memory, precision, batch size, and acceptance coverage. Fix class definitions and independent test units within each comparison. No generative-versus-direct latency result currently exists in this study.

Any post-hoc temperature or acceptance threshold is fitted on source development units for the target-free experiment. A different candidate set or target distribution does not inherit a calibration guarantee. Backbone independence is a method contract; empirical portability requires separately trained instances from distinct compatible families and a comparison against each family’s controls. Neither portability nor cross-dataset superiority follows from a successful import or forward pass.

4.5 Implemented instance and source-only transfer protocol

The implemented waveform study compares shared fault-description scoring with a fixed-class head on the same frozen backbone. Both sources supply exact labels over healthy, inner-ring damage and outer-ring damage; this study does not test a native coarse-label increment or a conditional-preservation intervention. The protocol below was fixed before external predictions. Implementation and source-interface acceptance are distinct from external diagnostic and latency evidence. Section 4.6 reports the source-fitting outcome and stopping boundary; external evaluation was not reached.

4.5.1 Population, measurement handling and evaluation units

The source population contains 29 documented PU bearings and 2,319 indexed recordings. KB23, KB24 and KB27 are outside this prospectively defined single-ring-component population. Their 240 recordings remain in the completed 32-bearing pilot and its mixed-damage counterexample; the new population neither repairs that mapping nor asserts microscopic single-defect purity. Fixed bearing roles are:

Source class	Train	Tune	Calibration	Internal test
Healthy	3	1	1	1
Inner-ring damage	6	2	1	2
Outer-ring damage	7	2	1	2
Total bearings	16	5	3	5
Original recordings	1,279	400	240	400

Recordings inherit their bearing's role. PU internal testing is exploratory because the earlier PU results informed development. The external MFPT population comprises all 20 known-label test-rig recordings: three baseline, ten outer-ring and seven inner-ring recordings. The publisher's example train/test folders both belong to this external population and provide no permission for target fitting or selection. No independently verified file-to-specimen map is available. Accordingly, the evaluation explicitly assumes one native MFPT experiment/recording per bearing, yielding 20 assumed experiment units. This convention is an assumption about the archive, not evidence that it contains 20 independent physical bearings or that a result generalizes to a machine population.

PHMFactory supplies the records, grouped source roles and windows. PU vibration channel index 2 and MFPT channel index 0 are selected explicitly. Format-only preservation of the processed H5 values is supplied provenance, rather than an independently reconstructed conversion history. Both sources retain native amplitudes with multiplier 1. PU's vibration amplitude unit is unspecified; the published MFPT `gs` convention is acceleration in g. There is therefore no claim of common-g calibration. The metadata nominal rate is 64,000 Hz for PU. Six MFPT recordings, comprising three healthy and three outer-ring records, have nominal rate 97,656 Hz; the other fourteen, seven inner-ring and seven outer-ring records, have nominal rate 48,828 Hz. Whole records are anti-alias resampled from these rates to 25.6 kHz using polyphase FIR processing with a Kaiser window ($\beta = 5$) and constant boundary extension. Each 8,192-point window spans 0.32 s at that rate. Windows are nonoverlapping, never cross a recording, and do not pad incomplete tails; discarded tails are retained in the export records. A global mean and standard deviation fitted only on PU training observations, with the record-within-unit weights above, scale the neural inputs. Raw sensor IDs,

filenames, original rates and lengths, source identity, split roles and true labels are not model inputs. Unverified cross-source amplitude calibration remains a transfer limitation. MFPT diagnosis and original acquisition rate are partly confounded; common-rate resampling does not certify identical acquisition chains.

4.5.2 Shared interface, controls and source-only fitting

The first backbone instance is frozen Qwen3-0.6B. A trainable linear adapter maps 64 contiguous patches of 128 samples into its input width; at the common rate each patch spans 5 ms. The causal decoder remains frozen while propagating gradients to the adapter. Signal hidden states are mean pooled. The same frozen decoder encodes the three fixed fault descriptions, with masked mean pooling and candidate vectors held fixed during fitting. A dot product with score temperature $\tau = 1$ produces the semantic logits. There is no additional learned candidate encoder, conditional KL term or projected update in this instance.

Family	Input/readout and comparison role
RF	Random forest on six time statistics and six Hilbert-envelope statistics from the common windows; practical numerical reference
Compact signal model	PHMFactory MobileNet1D, width multiplier 0.5 and dropout 0.2; trained signal reference
Indexed	Same patch adapter, frozen decoder and pooling; trainable three-class affine head
Semantic	Same signal interface; fixed aligned fault-description vectors and dot-product scores
Permuted semantic	Same vector set, parameter count and fitting procedure; fixed healthy-to-inner, inner-to-outer and outer-to-healthy description assignment throughout training and evaluation
Token score	Same signal observation and frozen backbone; all three verified label-token logits at the declared answer boundary, with complete class definitions in the prompt

The permuted control tests this fixed description-to-class assignment, including its interaction with fitting; it is not an inference-only corruption of a fitted model or proof of general language understanding. Exact token scoring and shortest legal one-token generation use the same selected token-model weights, prompt and label codes. Changing readout does not supply the token model with more measured signal.

Fitting uses PU train only; PU tune raw NLL selects configurations and checkpoints. Neural families use AdamW, batch size 16, learning rates $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}\}$ and weight decays $\{0, 10^{-2}\}$. Validation includes initialization and occurs every 100 updates, with five consecutive nonimprovements as the early-stop rule and hard limits of 2,000 updates and 30 minutes per candidate. The six settings use seed 101; the selected setting is refitted with seeds 202 and 303, without seed selection. RF uses 300 trees and the six combinations of maximum depth $\{8, \text{unlimited}\}$ and minimum leaf size $\{1, 4, 12\}$, with seed 101 and a 20-minute limit per candidate. The maximum matrix contains 40 neural fits and six RF fits; a configured ceiling is not a count of successful runs. A cap reached before qualification is retained as unresolved fitting adequacy

and does not authorize external target release. All source selections and checkpoint checks precede MFPT waveform prediction. One source-trained checkpoint per selected fit is reused for PU internal and MFPT external evaluation.

4.5.3 Endpoints, uncertainty and computational scope

The primary contrast is semantic-minus-indexed raw MFPT NLL under $R_{\mathrm{record/unit}}$. Error, Macro-F1 from the weighted confusion matrix, multiclass sum Brier and ten-bin equal-width ECE are secondary endpoints. The protocol calls for three selected fits per neural family and one for RF; RF's fit-to-fit SD cannot be estimated. Paired intervals resample declared evaluation units after averaging seed-specific differences within a unit. They condition on the saved fits and, for MFPT, on the assumed experiment independence; they do not cover all fitting, split or specimen uncertainty. The old pilot's bearing intervals are separate quantities.

Optional scalar temperature scaling uses only the PU calibration role, $T \in [\,0.05, 20\,]$, retaining $T = 1$ unless calibration NLL improves. Primary probabilities remain raw. Retrospective risk–coverage curves do not choose deployment thresholds; fixed fault-probability threshold 0.5 and declared acceptance-threshold sensitivity are reported separately. False alarms and missed faults include all eligible observations, including rejected diagnoses. No maintenance-cost interpretation is assigned to this study, and accepted error remains undefined at zero coverage.

Matched processing measurements use batch size one, declared precision, synchronization, 30 warmups and 200 observations per readout in five interleaved blocks. Full-coverage error is paired with the same selected fit used for timing. Warm common-rate-window processing, initial uncached candidate encoding on a resident backbone, model loading and whole-record preprocessing are separate measurements; their percentiles are not added. Acquisition duration is not processing latency. This single-backbone, three-candidate, fixed-token-budget study cannot establish empirical backbone portability or scaling with token budget or candidate count. Any efficacy or computational advantage remains conditional on valid completed outputs.

4.6 Source-side feasibility and stopping outcome

Real source-interface checks exercised the registered signal model and frozen Qwen weights, including adapter gradients and checkpoint reloads. The source-fitting stage produced 28 completed fit records: six RF trials, eight compact-signal fits, eight indexed-head fits and six semantic trials. For the compact-signal and indexed families, eight fits comprise six seed-101 configurations and two refits of the source-selected configuration. The semantic count comprises six configurations of one seed, not six independent replications. All NLL values below use the five PU tuning bearings and the record-within-unit weights; they are development selection scores, not held-out diagnostic performance.

Family	Selected setting	Tune NLL, seed 101	Tune NLL, seed 202	Tune NLL, seed 303	Source-fitting outcome
RF	Depth 8; minimum leaf size 12	1.041239	NA	NA	Completed source selection

Family	Selected setting	Tune NLL, seed 101	Tune NLL, seed 202	Tune NLL, seed 303	Source-fitting outcome
MobileNet1D	Learning rate 3×10^{-4}; decay 10^{-2}	0.662008	0.890223	0.631538	All three selected fits stopped by the source rule
Indexed	Learning rate 10^{-3}; decay 10^{-2}	0.714742	0.888396	0.775670	All three selected fits stopped by the source rule
Semantic	Learning rate 10^{-3}; decay 10^{-2}	8.738484	NA	NA	Update cap reached; fitting qualification unresolved

The six semantic tuning NLLs in fixed configuration order were 19.375280, 20.635522, 21.053214, 18.020516, 11.416944 and 8.738484. The winning configuration selected its checkpoint at update 1,700 but reached the 2,000-update cap without meeting the source early-stop rule. Its recorded qualification is therefore unresolved under the bounded protocol. The incomplete second-seed semantic refit was stopped at this boundary and supplies no completed replicate. The permuted-semantic and token-score families passed interface checks but were not fitted. Complete source-selection values and termination records are retained in the source-fit table.

The study stopped under the predeclared source-qualification rule before MFPT waveforms were opened. No external predictions, source-calibration results, paired inference intervals or matched latency results were obtained. The source scores show that this semantic configuration did not achieve qualified fitting within the stated search and update limits; they do not identify the cause, measure cross-dataset performance, or establish failure of semantic readout in general. No target-dependent repair, alternative seed selection or additional configuration search was performed. The exact-label transfer effect and native-granularity increment remain untested, while the completed compact-state pilot and separate simulation retain their original scope.

5. Completed Pilot: Experimental Protocol

The pilot tests whether changing and composing diagnostic questions helps a frozen compact-statistic predictor. It does not train a signal–description encoder, use coarse-source supervision, compare backbone families, or exclude an entire target dataset. Its detailed prediction and calibration equations are retained in Appendix C; the six fitted arm names below remain unchanged for reproducibility. Generic method statements in Section 4 must not be inferred from these arm names.

5.1 Dataset and physical grouping

We use PHM-Vibench's PU collection `RM_027_PU`, upstream dataset identifier 20. The existing H5 and canonical metadata contain 2,559 indexed recordings from 32 physical bearings: six healthy, 13 inner-ring-dominant, and 13 outer-ring-dominant. These are bearing counts, not file counts. All available indexed operating conditions and repetitions are retained.

The vibration sensor is upstream channel index 2. Metadata gives a nominal sampling rate of 64 kHz. Windows use actual stored sample counts without resampling; we do not independently establish each raw time field's association with the stored vibration channel. Durations below are nominal. No frequency-specific interpretation depends on an inferred exact timebase.

The source's dominant-category assignments include mixed damage in KB23, KB24, and KB27. They are retained with this limitation rather than relabeled as pure defects. The task is classification of the declared source categories, not multi-label localization. One raw record, `KA08/N15_M01_F10_KA08_2.mat`, is absent from the existing canonical metadata and H5 following an observed read failure. Its historical cause is unknown. No missing signal is synthesized and no user data are repaired in place.

5.2 Parameterized coverage and observation selection

Window length is selected from $W \in \{2048, 4096, 8192\}$. For maximum start interval S, with $1 \le S \le W$, and longest selected recording $L_{\max}$, the shared number of windows per recording is

$$N(W, S) = 1 + \left\lceil \frac{L_{\max} - W}{S} \right\rceil.$$

Upstream evenly spaced starts run from zero to $L_j - W$ in each recording. The first and last windows cover both ends, adjacent windows leave no gaps, and starts remain unique. A smaller S increases overlap. Default $S = W$ gives full coverage with the smallest shared count, replacing the fixed 64-position limit. Actual intervals, start gaps, and overlap are retained. A too-short record or failed coverage check causes an error instead of padding or silent exclusion.

For stored maximum length $L_{\max} = 299038$, candidate geometry is:

Window length	Nominal duration	Default maximum start gap	Windows per recording	Total windows
2,048	32 ms	2,048	147	376,173
4,096	64 ms	4,096	74	189,366
8,192	128 ms	8,192	37	94,683

These totals follow from coverage and the 2,559 indexed recordings; they are not independent sample sizes. Full coverage increases observed signal extent and often creates overlap. It does not imply exhaustive independent evidence or uniform sampling in physical time.

Window selection uses only the current split's train and tune bearings. Numerical-flat on amplitude-preserving statistics jointly searches window lengths and $C_{\text{reg}} \in \{0.01, 0.1, 1, 10, 100\}$ by tuning-bearing NLL. Exact ties prefer fewer windows, then smaller C_{reg}. Every arm shares the selected protocol in that split, including the standardized observation comparison. This choice favors the numerical baseline's observation criterion rather than an observation claimed optimal for all arms. Calibration and test predictions never choose window length or overlap. The existing configuration key remains `C`; the notation distinguishes it from the action-cost matrix.

5.3 Main split and repeated held-out bearings

The main split sorts bearings within each class, shuffles using seed 20260922, and assigns roles as follows:

Class	Train	Tune	Calibration	Test
Healthy	2	1	1	2
Inner-ring-dominant	7	1	2	3
Outer-ring-dominant	7	1	2	3
Total	16	3	5	8

Recordings and windows inherit their bearing's role. PHMFactory's validation role covers tune and calibration; the adapter distinguishes their whole-bearing assignments. Healthy bearings support all four roles but remain sparse, limiting claims about healthy-equipment diversity.

Six additional rotations provide a descriptive check across the healthy bearings. Each is tested once, the next in the frozen circular order calibrates, the following tunes, and the other three train. Each fault class is divided into six frozen test buckets of sizes 3, 2, 2, 2, 2, and 2. In a rotation, two remaining fault bearings calibrate, the next tunes, and the rest train. Every rotation repeats its own window and head selection using development bearings.

The first rotation has 17 training and seven test bearings; rotations 2–6 have 19 training and five test bearings. Each has three tuning and five calibration bearings. These rotations reuse the collection with overlapping fitting assignments and different training sizes and held-out compositions. They are not independent confirmations or pure perturbations of a fixed test set. The protocol is not revised between rotations based on test outcomes. Foldwise differences and their range describe this combined split sensitivity rather than six independent training seeds.

5.4 Comparison arms and fitting

The feasibility comparison crosses Numerical-flat and the original non-question-specific Qwen representation with raw, pooled, and equal-bearing calibration. It is reported for both frozen preprocessing protocols and is not relabeled as decomposition evidence.

The decomposition comparison contains six arms:

Arm	Representation and output	Purpose
Numerical-flat	Six measured values; one leaf head	Same-observation numerical reference
Numerical-factorized	Same values; presence and fault-only heads	Hierarchy without Qwen
Qwen-flat	Flat question A; one leaf head	Direct language reference
Qwen-shared-factorized	Flat A representation; presence and fault-only heads	Hierarchy on the direct language representation
Qwen-factorized	Presence and conditional-identity representations	Question-conditioned factorization
Qwen-two-question-direct	Flat A/B representations; two leaf heads	Additional-question control

Flat A and B share all category definitions and the healthy/fault grouping supplied to conditional questions. The two-direct-question control averages uncalibrated probabilities with fixed weights 1/2 before final calibration; no learned or test-selected fusion is used. One shared C_{reg} is selected at workflow level for every two-head method. All arms receive the same labeled recordings, window weights, five-value regularization search, tuning objective, and calibration bearings.

Heads use deterministic `lbfgs`, at most 2,000 iterations, and tolerance 10^{-7}, with convergence status retained. Each rotation refits training-only standardizers and heads, selects C_{reg}, and fits temperature. Frozen, protocol-compatible hidden states can be reused across role assignments without fitting to their labels. States from a different observation, interval, question, or model revision are not interchangeable.

The Qwen representation has 1,024 coordinates. Fitted-head parameter counts, including intercepts, are 3,075 for Qwen-flat, 2,050 for each Qwen factorized workflow, and 6,150 for the two-direct-question control. Numerical-flat has 21 parameters and Numerical-factorized 14. These counts make the different capacities and binary reference-logit parameterizations explicit rather than asserting exact model equivalence.

Pooled final-distribution temperature calibration is primary in the decomposition comparison. Raw and equal-bearing-scaled outputs are secondary analyses. All controls supply complete normalized distributions and the same leaf-or-defer contract. Equal question counts do not imply equal token counts, floating-point operations (FLOPs), latency, or parameter counts. We record tokens, padded tokens, forward batches, cache-extraction elapsed time, and fitted-head parameters. FLOPs are not measured, and unequal extraction schedules and warmup conditions do not constitute a matched deployment-latency benchmark.

5.5 Metrics and uncertainty

The primary endpoint is the paired bearing-weighted NLL difference, Qwen-factorized minus Qwen-flat; negative values favor factorization. Comparisons with Qwen-shared-factorized and Qwen-two-question-direct distinguish question conditioning from hierarchy and added evaluations. Numerical-factorized minus Numerical-flat tests whether the numerical hierarchy explains the effect's direction.

Multiclass Brier sums over all three classes, $\sum_k \left(p_k - \mathbb{1}\left[y = k \right] \right)^2$, rather than using a one-component binary convention. Accuracy and macro-averaged F1 (Macro-F1) use the same equal-bearing, equal-window weights as NLL. ECE uses maximum probability and ten bins: $[0, 0.1), [0.1, 0.2), \ldots, \left[0.9, 1 \right]$. It sums each bin's probability mass times the absolute difference between its weighted mean confidence and accuracy; empty bins contribute zero. ECE is descriptive and depends on binning.

Risk–coverage curves sort saved maximum probabilities with stable ties, and report weighted accepted-window error against weighted coverage. AURC is the sum of each prefix's risk times its coverage increment, using right endpoints; lower values indicate lower integrated selective error. Fixed-cost outputs include coverage, accepted-sample error when coverage is positive, and realized cost including every deferral's 0.25 charge. Lower selective error from greater rejection is not treated as improvement at matched coverage.

Always deferring is an analytical reference with zero coverage and cost 0.25. A model attaining that cost by rejecting every window has made no accepted diagnosis; cost and coverage must therefore be interpreted jointly.

For the main split, 2,000 paired bootstrap draws resample its eight test bearings using seed 20260922. Each draw uses the same bearings for the compared methods. Only Qwen-factorized minus Qwen-flat is the prespecified primary contrast. The other reported 95% intervals are descriptive, unadjusted for multiple

comparisons. All intervals condition on the split and fitted models, omitting training and selection variation. Overlapped windows are never bootstrap units. Foldwise differences are separate because fitting data overlap across rotations. Failure to exclude zero is not evidence of equivalence.

5.6 Reproducibility and predeclared interpretation

Artifacts retain grouped roles, actual intervals, observations and weights, queries, dependency and model versions, fitted standardizers and heads, tuning values, temperatures, raw factors, final probabilities, per-bearing losses, and failures. Figures and additional cost analyses read saved predictions without rerunning Qwen. Encoding records sequence lengths, tokens, padded tokens, forward batches, precision, hardware, and cache-extraction elapsed time. These describe actual resource use without estimating FLOPs or establishing deployment latency; no proprietary Jev speed ratio or end-to-end maintenance speed claim is imported.

For the complete 376,173-window amplitude-preserving 2,048-point export, the direct flat query processes 64,870,462 non-padding tokens in 1,470 batches. The presence and identity queries together process 135,759,692 tokens in 2,940 batches; the two-direct-query control processes 131,621,789 tokens in 2,940 batches. Corresponding padded-token totals are 65,199,017, 136,416,802, and 132,278,899. Thus two questions do not imply identical token cost. Shared-representation factorization reuses the flat query without another encoder call. These complete-cohort cache totals include all roles and are reused across protocols selecting that geometry; they are not training-repeat counts or deployment latency measurements. Per-query elapsed extraction times and the other geometry's counts are retained in the encoding table.

A numerical hierarchy gain supports a hierarchy explanation. A gain retained by Qwen-shared-factorized does not establish that two diagnostic questions are necessary. A gain matched by two direct questions is compatible with extra calls or prompt diversity. A gain following amplitude restoration belongs to the observation revision unless within-protocol controls isolate an additional effect. Zero and negative results remain findings. Interface or infrastructure failure is separated from a completed comparison.

6. Completed Pilot: Results

6.1 Observation and coverage verification

The exported 2,048-point protocol covers every stored sample in all 2,559 recordings, yielding 376,173 windows. Actual adjacent start gaps range from 1,697 to 2,035 samples, corresponding to overlaps of 13–351 samples. In the per-window-standardized export, RMS and standard deviation both have mean 0.9999999972 and across-window standard deviation 6.58×10^{-7}, with values ranging from 0.99999442 to 1.00000485. They are nearly constant rather than exactly constant: training standardization rescales the remaining finite-precision variation, so these observations do not establish an exactly reduced-dimensional state. The amplitude-preserving export remains a separately frozen comparison. These are input and coverage measurements, not diagnostic-performance results.

6.2 Amplitude-preserving feasibility comparison

The completed amplitude-preserving feasibility run uses the selected 2,048-point window on the main split. It evaluates 94,080 test windows from eight bearings, with 11,760 windows per bearing. Its five calibration bearings provide 58,653 windows: four have 11,760 and KA08 has 11,613, reflecting the existing missing record. The numerical and legacy-Qwen heads both select $C_{\mathrm{reg}} = 0.01$ on the tuning bearings. Qwen encoding uses frozen bfloat16 weights; this comparison retains the non-question-specific legacy representation rather than the fixed diagnostic questions.

Representation	Calibration	Test NLL	Brier	ECE	Coverage (%)	Cost
Numerical	None	1.0708	0.6555	0.1208	13.5406	0.2850
Numerical	Pooled	1.0227	0.6216	0.1110	0.0744	0.2498
Numerical	Equal-bearing	1.0225	0.6215	0.1105	0.0776	0.2498
Legacy Qwen	None	1.2595	0.7145	0.2008	35.8578	0.3461
Legacy Qwen	Pooled	1.0527	0.6370	0.0818	0	0.2500
Legacy Qwen	Equal-bearing	1.0526	0.6369	0.0816	0	0.2500

All values use equal-bearing evaluation weights; Brier sums the three class components. Numerical accuracy is 45.7345% and Macro-F1 is 0.3484; legacy-Qwen accuracy is 47.4596% and Macro-F1 is 0.4086. Each pair of class metrics is unchanged across its temperature variants, as required by global positive scaling. The pooled temperatures are 4.4970 for numerical and 7.2199 for Qwen; equal-bearing temperatures are 4.4776 and 7.1820. Pooled calibration lowers observed test NLL in both representations, while the calibrated legacy-Qwen NLL remains higher than its numerical counterpart. This is representation/calibration evidence, not a decomposition result.

Equal-bearing minus pooled NLL is -0.000171 for numerical, with conditional paired 95% interval $[-0.000557, 0.000270]$, and -0.000162 for legacy Qwen, with interval $[-0.000597, 0.000319]$. Both intervals include zero; these data do not establish an advantage or equivalence. The calibration counts differ only by one recording, so this collection provides little contrast between the two weighting populations.

The uncalibrated accepted-window error is 50.8125% for numerical and 51.8127% for legacy Qwen, yielding costs above the always-defer reference. Pooled numerical calibration accepts only 70 of the 94,080 windows, with no observed errors among those accepted; equal-bearing calibration accepts 73, also without observed errors. Both calibrated Qwen variants defer every window. Their cost of 0.25 therefore reflects zero diagnostic coverage, and the numerical cost near 0.25 reflects almost complete deferral rather than broad reliable diagnosis.

6.3 Standardized-observation feasibility comparison

The completed per-window-standardized feasibility run uses the same window intervals, eight test bearings, 94,080 test windows, labels, and role assignments. Both heads again select $C_{\mathrm{reg}} = 0.01$. It tests the original observation protocol, including the nearly constant RMS/standard-deviation coordinates and their rescaled finite-precision variation described above.

Representation	Calibration	Test NLL	Brier	ECE	Coverage (%)	Cost
Numerical	None	1.2616	0.7118	0.1215	2.2704	0.2668
Numerical	Pooled	1.0961	0.6647	0.0791	0	0.2500
Numerical	Equal-bearing	1.0961	0.6647	0.0796	0	0.2500
Legacy Qwen	None	1.1656	0.6975	0.1349	0.0797	0.2504

Representation	Calibration	Test NLL	Brier	ECE	Coverage (%)	Cost
Legacy Qwen	Pooled	1.0897	0.6618	0.0525	0	0.2500
Legacy Qwen	Equal-bearing	1.0898	0.6619	0.0528	0	0.2500

Numerical accuracy is 43.1186% and Macro-F1 is 0.3277; legacy-Qwen accuracy is 37.9188% and Macro-F1 is 0.2934. These class metrics remain invariant under temperature scaling. Numerical pooled and equal-bearing temperatures are 8.6600 and 8.5583; the corresponding Qwen temperatures are 1.9971 and 1.9916. Both calibrated representations defer all test windows, attaining the always-defer cost and leaving accepted-sample error undefined.

Equal-bearing minus pooled NLL is 0.000012 for numerical, with conditional paired 95% interval $[-0.000605, 0.000751]$, and 0.000108 for legacy Qwen, with interval $[-0.000329, 0.000581]$. These small, uncertain differences do not establish a weighting advantage. Across the separately frozen feasibility protocols, preserving amplitudes reduces pooled test NLL by 0.0734 for numerical and 0.0370 for legacy Qwen. This observed preprocessing contrast supports retaining the distinction between observation and classifier effects; it is not a finding about question decomposition or broader industrial generalization.

6.4 Primary decomposition comparison

All six decomposition arms completed under the frozen amplitude-preserving observation and the same 94,080 test windows from eight held-out bearings. Every workflow selects $C_{\mathrm{reg}} = 0.01$; calibration acts on each final three-class distribution. The numerical-flat result is the protocol-compatible feasibility result, retained as a shared comparison rather than independent evidence.

Method	NLL	Brier	Accuracy (%)	Macro-F1	ECE	AURC
Numerical-flat	1.0227	0.6216	45.7345	0.3484	0.1110	0.4721
Numerical-factorized	1.0183	0.6190	45.9067	0.3496	0.1055	0.4642
Qwen-flat	1.0553	0.6386	46.4605	0.3884	0.0760	0.4934
Qwen-shared-factorized	1.0463	0.6334	46.3520	0.3859	0.0639	0.4915
Qwen-factorized	1.0542	0.6387	45.9460	0.3782	0.0650	0.5399
Qwen-two-question-direct	1.0568	0.6397	47.2523	0.3905	0.0854	0.5089

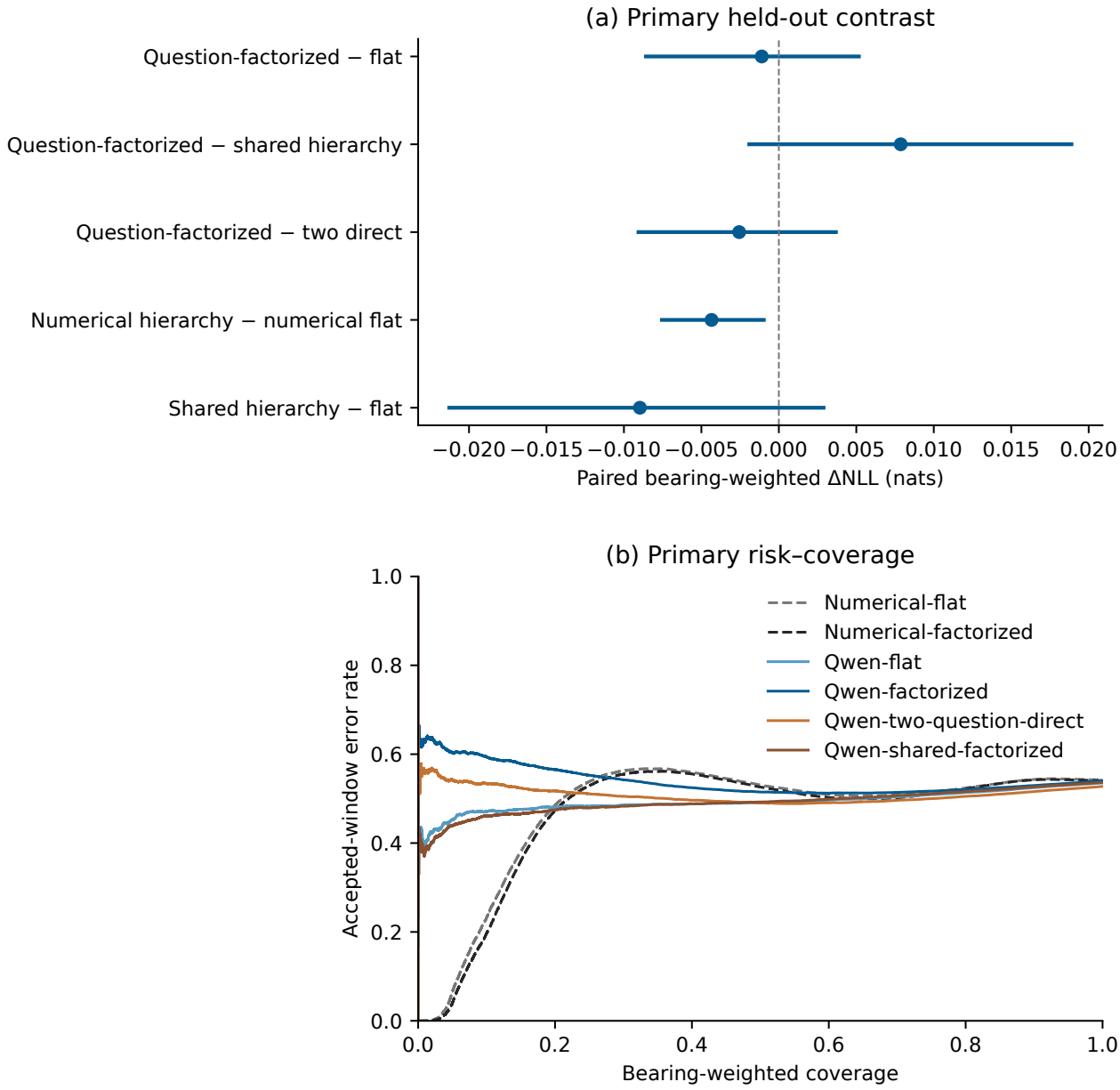
The primary Qwen-factorized-minus-Qwen-flat NLL difference is −0.001102, with conditional paired 95% interval [−0.008577, 0.005163]. Against the shared-representation factorization, the difference is +0.007866 [−0.001909, 0.018887]; against two direct questions it is −0.002561 [−0.009060, 0.003682]. These intervals

do not establish an additional question-conditioning benefit or equivalence. The shared-representation hierarchy has lower point NLL than the question-conditioned model without a second encoding.

Numerical factorization has the lowest primary NLL. Its paired difference from Numerical-flat is −0.004344 [−0.007547, −0.000973]. This secondary, unadjusted conditional interval excludes zero, but describes an ordinary hierarchy on this split and fit; it is neither a language-model contribution nor independent confirmation across populations. The shared-Qwen-hierarchy-minus-flat difference is −0.008968 [−0.021275, 0.002886], retaining uncertainty about that descriptive contrast.

Probability and selection metrics disagree in an informative way. Qwen-factorized has ECE 0.0650 versus Qwen-flat's 0.0760, yet AURC is higher, 0.5399 versus 0.4934. Its Brier and class metrics also do not improve over Qwen-flat. Thus the lower ECE does not establish better confidence ordering or overall diagnostic utility. At the fixed cost threshold, all four calibrated Qwen methods have zero coverage and cost 0.25. Numerical-flat accepts 70 windows and Numerical-factorized 25, both without observed errors among those accepted; their costs are 0.249814 and 0.249934. These almost-complete deferrals are interpreted relative to always deferring, rather than as useful broad diagnostic coverage.

Before final calibration, Qwen-factorized NLL is 1.2429 versus 1.2237 for Qwen-flat, and its cost is 0.3511 versus 0.3311. The tiny favorable calibrated primary point difference therefore does not persist in the uncalibrated comparison. Raw and equal-bearing-calibrated probabilities, class metrics, and risk–coverage curves are retained in the complete result tables rather than used to select a favorable calibration regime.



Completed pilot only: main-split paired bearing NLL differences and pooled-calibration risk–coverage curves from saved predictions. Only Qwen-factorized minus Qwen-flat is the prespecified primary contrast; other conditional 95% intervals are descriptive and unadjusted. These are not tests of semantic candidate learning or cross-dataset generalization.

6.5 Split-sensitive attribution

All six predeclared rotations completed. Rotation 5 selects 2,048-point windows; the other five select 8,192 points using their own development bearings. Every method within a rotation shares its selected observation. The paired bearing-weighted NLL differences below are descriptive because fitting and evaluation populations overlap across rotations. Negative values favor question-conditioned factorization.

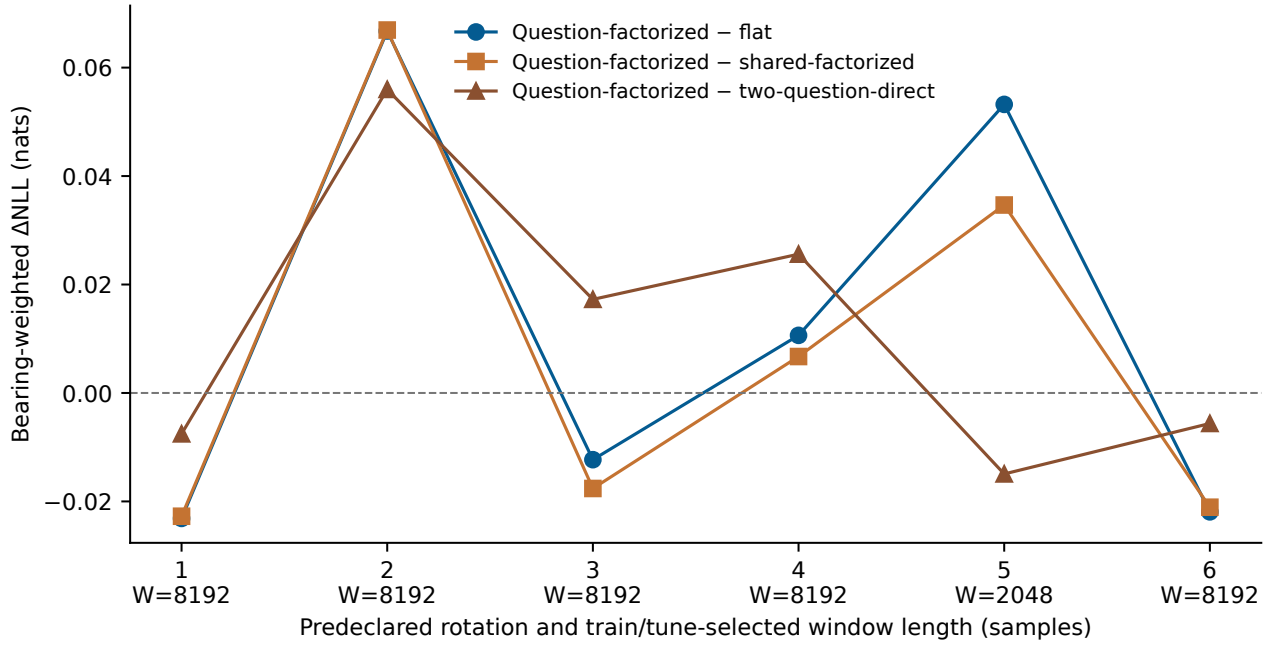
Rotation	Window length	Against Qwen-flat	Against shared-representation factorization	Against two direct questions
1	8,192	−0.0231	−0.0227	−0.0075
2	8,192	+0.0667	+0.0669	+0.0560

Rotation	Window length	Against Qwen-flat	Against shared-representation factorization	Against two direct questions
3	8,192	−0.0123	−0.0176	+0.0173
4	8,192	+0.0106	+0.0067	+0.0256
5	2,048	+0.0532	+0.0347	−0.0149
6	8,192	−0.0219	−0.0211	−0.0056

Each contrast favors factorization in three rotations and disfavors it in three. Against Qwen-flat, differences range from −0.0231 to +0.0667; against two direct questions, from −0.0149 to +0.0560. In rotation 3, factorization improves NLL relative to the flat question while two direct questions perform better than factorization. In rotation 5, factorization improves on two direct questions while being worse than the flat and shared-representation hierarchy. These reversals do not support a stable question-conditioning increment.

Per-bearing values reveal heterogeneous changes. In rotation 2, healthy bearing K002's factorized-minus-flat difference is +0.4008, contributing +0.0802 to the five-bearing mean of +0.0667. In rotation 5, K004's difference is +0.2535, contributing +0.0507 to the mean of +0.0532. In rotation 4, KA01's difference is +0.1381 despite three other bearings having negative differences. These are descriptions of error concentration, not evidence for a physical mechanism or independent replication.

Higher acceptance also need not improve the fixed-cost decision. In rotation 5, Qwen-factorized has 5.1395% coverage and cost 0.2808; Qwen-flat has 4.1446% coverage and cost 0.2739. Both exceed the 0.25 always-defer reference. Thus accepting more windows does not by itself demonstrate useful diagnosis. All rotation probabilities and metrics, including negative results and the numerical hierarchy controls, are retained in the complete saved tables.



Completed pilot only: question-conditioned factorization minus three Qwen controls in six predeclared bearing rotations. Each rotation shares its development-selected window within the comparison. Direction changes describe sensitivity within one collection, not independent replications or target-dataset transfer.

7. Discussion and Required Validation

7.1 What the pilot establishes

The pilot does not establish an additional question-conditioning benefit. The shared-representation hierarchy has a better primary point NLL, and the question-conditioned contrasts change direction across refits. Because observations and model-selection opportunities are matched within a comparison, a favorable split cannot be selected as a general benefit. These conclusions apply to the frozen compact-statistic model, not to the proposed signal–semantic learner. No new model inference or cross-dataset result is introduced by the formulation.

Six temporal statistics discard waveform order and frequency structure. A pretrained encoder can impose a useful prior or reparameterize these statistics, but cannot retrieve the particular signal information that was not supplied. The pilot does not determine whether richer measured inputs or a trained signal adapter would improve diagnosis. Its few healthy bearings, source-dominant categories including mixed damage, and overlapping rotations limit external validity. Full signal coverage does not resolve limited equipment diversity.

The observed concentration of errors on particular bearings is descriptive, not a physical mechanism. The original protocol was frozen before these test observations; neither its questions nor feature definitions were selected to repair those outcomes. Hypotheses motivated by these errors are exploratory and require evaluation on untouched data. The new generalization questions arose after the pilot and are not retrospectively described as its preregistered objectives.

7.2 Validation of annotation-compatible transfer

The first pending comparison holds observations, backbone, descriptions, and source model-selection opportunity fixed while changing valid supervision. Compare compatible-event training with exact-label-only training and a justified common-coarse-label reference. Fine-label evaluation uses independently available ground truth. The coarse reference is assessed on common aggregated events; fine-state scores require an explicitly justified fine-state predictor. An inner-versus-outer task conditional on a known fault is reported separately from a task including healthy bearings. Neither conditional task selection nor coarser requested answers may give the proposed method an easier test population.

Exclude whole target datasets and dependent physical units from task-specific fitting and calibration, and document their membership in any domain-pretraining corpus. Target-adapted models and source-only models occupy separately identified protocols. Audit native class definitions before constructing S_d; do not turn run-to-failure timestamps into fault-onset labels without supporting annotations. Sources whose dominant or mixed faults cannot share the proposed partition are reported as incompatible rather than discarded after inspecting test performance. Controlled coarsening of source labels can diagnose sensitivity, but is not a substitute for demonstrating that the native annotations present the claimed problem. Compare results per target dataset, retaining negative transfer and uncertainty at the physical-unit level.

A conventional semantic matcher receiving identical supervision, architecture and optimization is an implementation-equivalence reference: no incremental algorithm has then been specified. The supervision contrast and comparison with a mapped shared classifier test different changes. A compact signal-only model tests whether the pretrained backbone adds value. Practical comparisons allow each method reasonable source-only tuning; mechanism comparisons hold the relevant inputs, supervision and adaptation

opportunities fixed. A single checkpoint is shared within each source-trained model instance; a different excluded-target fold generally trains a different checkpoint. Two compatible backbone families test method portability without implying universal performance.

7.3 Probability estimates, rejection, and runtime

The pilot's lower ECE but higher AURC distinguishes calibration from selective ordering. Complete deferral attains the always-defer cost but produces no accepted diagnosis. Calibration bearings are few, and nearly equal counts make the two calibration-weighting objectives very similar; their uncertain differences establish neither a weighting advantage nor equivalence. The normalized deferral cost does not model expert accuracy, delay, downtime, or physical intervention.

The second pending comparison measures diagnostic error or NLL jointly with latency while varying the declared signal-token budget and candidate count. Use the same hardware, precision, batching, warmup and input duration, and include device synchronization. Include the exact label-token control of Section 4.4, not only minimal-output generation, and compare an efficient numerical classifier. Separate prompt construction, backbone encoding, output projection, candidate compilation and device transfer when explaining a timing difference; zero decoded response tokens is not an adequate explanation by itself. Report cold candidate compilation separately from repeated warm decisions, and retain failures and format errors in the generated-output denominator. A method that trades away accuracy or rejects everything cannot be declared uniformly faster at the same diagnostic service level.

Acquisition duration is not inference latency, candidate order is not class semantics, and a source-calibrated maximum probability is not an engineering interlock. These distinctions constrain interpretation but do not guarantee the proposed method's outcome. The study contains no matched latency experiment or real target-dataset exclusion experiment for the semantic method. Section 7.4 reports a separate simulated coarse-label learning comparison, not evidence of native industrial coarsening or measured deployment speed.

7.4 Simulation qualification of conditional preservation

A separate, completed simulation uses 46 fixed signal features: six temporal statistics and 40 squared-envelope order-energy bins. Three independent cohorts contain 270 target virtual units, each contributing four windows. Fine-source labels distinguish healthy, inner-ring and outer-ring states; coarse-source labels distinguish healthy from faulty. Fixed impulse-response, modulation, noise and transfer conditions generate the waveforms. This is a controlled coarse-supervision experiment, not measured equipment data or an implementation of the semantic encoder in Section 4. Its complete equations, design parameters, source-only search and failures are retained in the simulation report, with computed results.

The soft control penalizes divergence from the initialized within-fault conditional distribution; zero penalty is eligible. A hard control instead adds the same residual to the two fault logits, retaining the initial group-mass offset and exactly preserving their log probability ratio. Ordinary joint event learning and the released weighted Libra loss provide matched-input references; a random forest trained only on fine-source labels is a different-information practical reference. All selected free-head extensions choose update step zero, and the soft control chooses zero penalty. Consequently, fine-only, joint-event, weighted-Libra and soft-anchor workflows have identical selected probabilities: mean target NLL 0.002140 and accuracy 99.907%. Their zero paired difference is caused by selection of the same predictor, not equivalence of active algorithms. Hard freezing yields NLL 0.001321 with unchanged accuracy; its smaller parameterization and penalty geometry are not matched to the full heads.

Positive-penalty trial paths reduce conditional drift, but those checkpoints are not selected. The source-side selection criterion is already near saturation at initialization, leaving the proposed soft-preservation benefit unresolved. Under the predeclared combined speed, slip and transfer mismatch, the common selected linear predictor falls to 40.370% accuracy; raw-spectrum-magnitude-preserving phase randomization yields 33.333%. These jointly specified stresses do not isolate an individual failure cause. The results motivate a source-only diagnostic of which fine distinctions remain unlearned, rather than changing target conditions to favor the intervention. No runtime advantage, real-domain transfer, or novel conditional-regularization principle follows from this simulation.

8. Conclusion

Jev-PHM is formulated as non-generative semantic fault-candidate prediction with compatible source supervision, independently of a particular backbone. The scientific questions concern the transfer of diagnostic distinctions under annotation and measurement changes, and the error–latency consequences of replacing response generation with direct scoring. Existing semantic classification and coarse-label theory delimit the method’s novelty. The native mixed-damage example shows why the annotated event must be determined by the chosen diagnostic state, while coarse supervision may still leave finer probabilities unidentified. Exact single-token label scoring further restricts the expected efficiency increment: both compared decisions can avoid sequential response decoding. These are applicability and comparison conditions, not demonstrated transfer or latency gains.

The completed 32-bearing pilot supplies a bounded result: two diagnostic questions do not demonstrate a stable increment over direct prediction, shared-representation hierarchy, or two direct questions in the six-statistic frozen-encoder setting. The primary NLL interval includes zero, rotation contrasts change sign, and primary calibrated Qwen methods defer all observations. These results remain intact. The separate simulation likewise does not establish an active soft-preservation gain: source selection returns the initial predictor, and declared mismatch produces substantial failure. This distinguishes an implemented constraint, its effect during a trial, and the behavior of the selected model. The waveform interface is implemented, but source-side fitting stopped before a semantic predictor qualified, leaving external efficacy, backbone portability and matched runtime effects untested.

Appendix A. Fixed pilot question meanings

Every decomposition prompt begins with this identical definition, including the actual label strings and mixed-damage qualification:

> The mutually exclusive classes are healthy, inner-race-dominant, outer-race-dominant. The healthy class is healthy; the defined fault classes are inner-race-dominant, outer-race-dominant. A dominant fault label identifies the predominant annotated damage location and may include mixed damage; it does not assert a pure isolated fault.

The fixed questions below precede the identical state:

Query	Question
Flat A	Which one of the defined classes describes this vibration window?

Query	Question
Flat B	Choose the most appropriate class from the complete defined diagnosis classes for this vibration window.
Presence	Does this vibration window belong to any of the defined fault classes rather than the healthy class?
Conditional identity	Conditional on this vibration window belonging to one of the defined fault classes, which fault class describes it?

The feasibility legacy query is “Represent this vibration window for fault diagnosis.” It lacks the category-conditioned diagnostic question and is analyzed only as the retained feasibility representation. No prompt contains a sample’s answer or identifier. Complete templates, tokenization settings, and class order are saved. These literal prompt strings are preserved; their use of “race” does not relabel the source categories or imply an additional damage class.

Appendix B. Data and software availability

The implementation and compact pilot summaries are maintained at PT01-jev-phm. PHMFactory/PHM-Vibench is fixed to commit `cdc0669167ca25ac848e34c880875ca39c6aed89` (PHMbench 2026). Paderborn source data remain subject to the provider’s CC BY-NC 4.0 terms, distinct from the original paper and code licenses. Large observations and hidden states are stored separately from version-controlled code.

The successful pilot run is `run-20261013-pu-001`. It contains two feasibility evaluations, the primary decomposition comparison, and six bearing rotations under `src/outputs/jev-phm/`. Each evaluation retains `metrics.json`, `predictions.npz`, configuration, and provenance. The compact study summary retains all evaluation metrics and fitted configuration, while per-bearing NLL values support numerical reconstruction of paired contrasts. Figures are generated by `src/S03_Scripts/plot_jev_results.py` from saved predictions. The repository reproduction instructions provide the pilot preparation, export, window selection, encoding, and evaluation commands using `src/S03_Scripts/run_jev_study.py`. They are not an executable implementation of Section 4. Reproduction uses a new output directory and retains failures.

The waveform source-feasibility study in Sections 4.5–4.6 is `run-real-20261019-001`, using the six configurations in `src/S02_Configs/real/` through the same study entry. Small per-fit result files, configurations and the source summary are retained with the repository; large observations and checkpoints remain in separate local study storage. The execution instructions distinguish reproduction and inspection of this stopped source study from an external evaluation, which was not performed. MFPT’s published examples use CC BY-NC-SA 4.0 terms; its assumed experiment-unit grouping does not establish a physical specimen roster.

The simulation discussed in Section 7.4 is the repository study `outputs/run-sim-002`, specified by `src/S02_Configs/jev_simulation_poc.json` and implemented by `src/S01_Package/jev_simulation.py`. Its committed report and compact CSV are linked above; saved native outputs and search histories belong to that run. A distinct earlier 16-feature simulation package is not pooled with this 46-feature experiment or the PU pilot. CWRU is excluded from all new experimental work in this project.

The sole authoritative manuscript is Markdown. `src/S03_Scripts/build_jev_paper.py` builds derived offline HTML and PDF previews without maintaining a second text source. Author identities, funding, conflicts, target journal, and submission declarations remain unconfirmed. The complete previous candidate and its provenance are retained in Git at `ff9db6e4e43c8da2fc66990a03dcc78ed1fd2afb`.

Appendix C. Completed pilot prediction and estimation details

C.1 Scope, features, and frozen encoder

The pilot has $\mathcal{C} = \{H, I, O\}$: healthy, inner-ring-dominant, and outer-ring-dominant states. Every recording from a bearing remains in one fitting or evaluation role. Labels, bearing/file IDs, operating-condition IDs and role assignments are excluded from inputs. PHMFactory owns reading, preprocessing, window construction, and grouped assignments; validation bearings are divided into tuning and calibration roles without changing the original data.

For a univariate window of length W, the feature vector is

$$s(x) = \left(\sqrt{\bar{x^2}}, \sqrt{(x - \bar{x})^2}, |\bar{x}|, \max x - \min x, \frac{\max |x|}{\sqrt{\bar{x^2}}}, \left(\frac{x - \bar{x}}{\operatorname{std}(x)} \right)^4 \right).$$

These are root mean square (RMS) amplitude, population standard deviation, mean absolute amplitude, peak-to-peak amplitude, crest factor, and ordinary kurtosis rather than excess kurtosis. A zero denominator gives the relevant ratio a declared zero convention. Values use fixed order and eight significant digits. The model does not calculate signal quantities. The two frozen preprocessing conditions are upstream per-window standardization and no signal normalization. They are shared by all arms within a comparison; the decomposition study uses amplitude-preserving observations. These choices were fixed before test-performance inspection, not selected to improve a test result.

Qwen/Qwen3-0.6B is the pilot instance, fixed to revision `c1899de289a04d12100db370d81485cdf75e47ca` (Qwen Team 2025). A non-thinking chat template supplies the common class definitions, a fixed question q, and the serialized statistics. The last non-padding hidden state $h_q(s) \in \mathbb{R}^{1024}$ is extracted without answer-token generation; the backbone is frozen. Training-only weighted standardization precedes each logistic head. The conditional-identity standardizer uses faulty training observations only. Numerical heads use the same six values without the encoder. These are supervised probes, not zero-shot diagnostic models.

C.2 Conditional heads and output probabilities

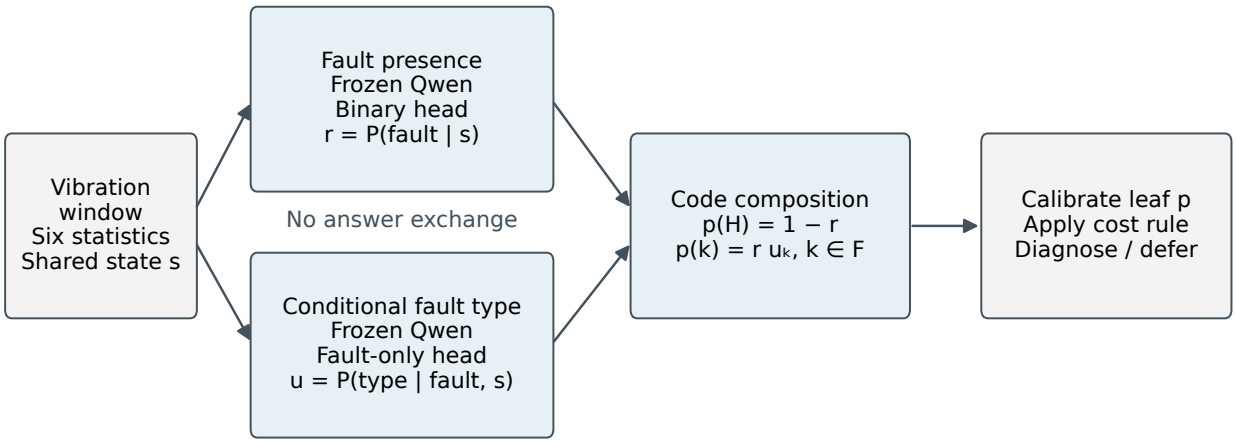
Let $F = (I, O)$. The pilot's presence probability is denoted $r(s)$. With sigmoid σ, learned coefficient vectors $\beta_F, \beta_L \in \mathbb{R}^{1024}$ and scalar intercepts b_F, b_L,

$$r(s) = \sigma\left(\beta_F^\top h_{q_F}(s) + b_F\right), \qquad u(s) = \operatorname{softmax}\begin{pmatrix} 0 \\ \beta_L^\top h_{q_L}(s) + b_L \end{pmatrix}.$$

I is the identity reference class and O the learned non-reference class. Each binary branch therefore has one learned coefficient vector and intercept, whereas the flat head learns a three-row multinomial matrix. Their regularization geometry differs even for equal C_{reg}. Code forms

$$p(H \mid s) = 1 - r(s), \qquad p(k \mid s) = r(s)\, u_k(s), \quad k \in F.$$

Presence targets are $\mathbb{1}[y \in F]$; identity targets exist only for faulty training observations. Both are derived from the same diagnostic label, not additional physical-symptom annotations. Every test observation executes both question branches; neither answer nor a test label is used to select the other branch. The figure's labels r and u denote these pilot factors, not the proposed semantic scorer.



Diagnosis-derived labels, not independently measured physical-concept supervision.

Completed pilot only: independent fault-presence and conditional fault-identity questions read the same six measured statistics. Diagnostic labels supervise the heads; code composes leaf probabilities, calibrates them, and applies a diagnosis/deferral rule. The figure notation r denotes the pilot fault probability. This diagram does not depict the new method in Section 4.

C.3 Training weights, loss, and regularization

A recording j with m_j exported windows assigns scalar training weight $w_{ji} = 64/m_j$. Its total training mass is always 64, preserving the earlier probe's loss scale rather than defining prevalence or an independent sample count. Fault-only heads retain the original weights without renormalizing them to the presence branch. The objective is

$$\mathcal{L}_{C_{\text{reg}}} = \sum_{i \in \text{train}} w_i \, \ell_{F,i} + \sum_{i \in \text{train}:\, y_i \in F} w_i \, \ell_{L,i} + \frac{\| \beta_F \|^2 + \| \beta_L \|^2}{2C_{\text{reg}}}.$$

Intercepts are unpenalized and class balancing is disabled. The data-fit term equals weighted leaf NLL because

$$-\log p\left(y_i \mid s_i\right) = \begin{cases} -\log\,(1 - r_i), & y_i = H, \\ -\log r_i - \log u_{i,y_i}, & y_i \in F. \end{cases}$$

Separately averaging and equally adding the two losses would generally change faulty-example weight and is not the implemented objective. Flat heads use corresponding weighted categorical NLL with L2 regularization. One shared C_{reg} is selected for each pair, not a Cartesian product of penalties. The regularization parameter is denoted C_{reg} to distinguish it from the action-cost matrix Λ.

C.4 Calibration population and temperature

Each method first forms its complete three-class distribution. A scalar temperature is fitted only on calibration bearings:

$$\tilde{p}_T(k \mid s) = \frac{p(k \mid s)^{1/T}}{\sum_{j \in \{H,I,O\}} p(j \mid s)^{1/T}}.$$

The search interval is $[0.05, 20]$; $T = 1$ is retained if the fitted candidate does not improve the selected calibration objective. Boundary solutions are reported. No extra node-specific calibration is used. Calibrated factors, if exposed, are reconstructed as $\tilde{r} = 1 - \tilde{p}_T(H)$ and $\tilde{u}_k = \tilde{p}_T(k) / \tilde{r}$, separately from raw factors.

For calibration bearing b, let $L_b(T)$ be mean window NLL and n_b its window count. Pooled calibration minimizes $\sum_b n_b L_b(T) / \sum_b n_b$, while equal-bearing calibration minimizes $B^{-1} \sum_b L_b(T)$. With covariance computed using divisor B and mean count $\bar{n}$,

$$R_{\text{window}}(T) - R_{\text{unit}}(T) = \frac{\text{Cov}_b(n_b, L_b(T))}{\bar{n}}.$$

This elementary identity does not predict held-out benefit. Equal counts make the objectives equal, and uniformly replicating all windows of one bearing leaves its equal-bearing objective unchanged. The weighting defines a population, not an arbitrary-distribution calibration guarantee.

Positive global temperature preserves class argmax and therefore each arm's accuracy and Macro-F1. In binary classification, maximum probability is $\sigma(|z_1 - z_0|/T)$, so confidence ordering and risk at matched coverage are unchanged under a fixed tie convention. No corresponding unconditional ordering claim holds for multiclass maximum probability. NLL, fixed-threshold acceptance and realized scenario cost can still change.

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